



KNN algo

```
In [99]: import pandas as pd
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
import matplotlib.pyplot as plt
from sklearn.neighbors import KNeighborsClassifier
```

```
In [100]: df= load_iris()
```

```
In [101]: x= df.data[:,1:4]
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In [102]: y= df.target[:,1:4]
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In [103]: x_train , x_test, y_train , y_test= train_test_split(x,y, test_size= 0.2, rand
```

```
In [104]: model = KNeighborsClassifier(n_neighbors=15, metric="minkowski")
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In [105]: model.fit(x_train, y_train)
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```
Out[105]: ▼ KNeighborsClassifier ⓘ ?
```

Parameters

 <code>n_neighbors</code>	15
 <code>weights</code>	'uniform'
 <code>algorithm</code>	'auto'
 <code>leaf_size</code>	30
 <code>p</code>	2
 <code>metric</code>	'minkowski'
 <code>metric_params</code>	None
 <code>n_jobs</code>	None

```
In [106]: model.score(x_test, y_test)
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Out[106]: 0.9545454545454546
```

```
In [107]: model.score(x_train, y_train)
```

```
Out[107]: 0.9545454545454546
```

```
In [108]: x_train[45]
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Out[108]: array([7.6, 3. , 6.6, 2.1])
```

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In [109... y_train[45]
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Out[109... np.int64(2)
```

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In [110... model.predict([[5.4, 3, 4.5,1.5]])
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Out[110... array([1])
```

```
In [111... from sklearn.metrics import classification_report, confusion_matrix
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```
In [112... y_pred= model.predict(x_train)
```

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In [113... print(classification_report(y_train, y_pred))
```

	precision	recall	f1-score	support
0	1.00	1.00	1.00	40
1	0.91	1.00	0.95	39
2	1.00	0.56	0.71	9
accuracy			0.95	88
macro avg	0.97	0.85	0.89	88
weighted avg	0.96	0.95	0.95	88

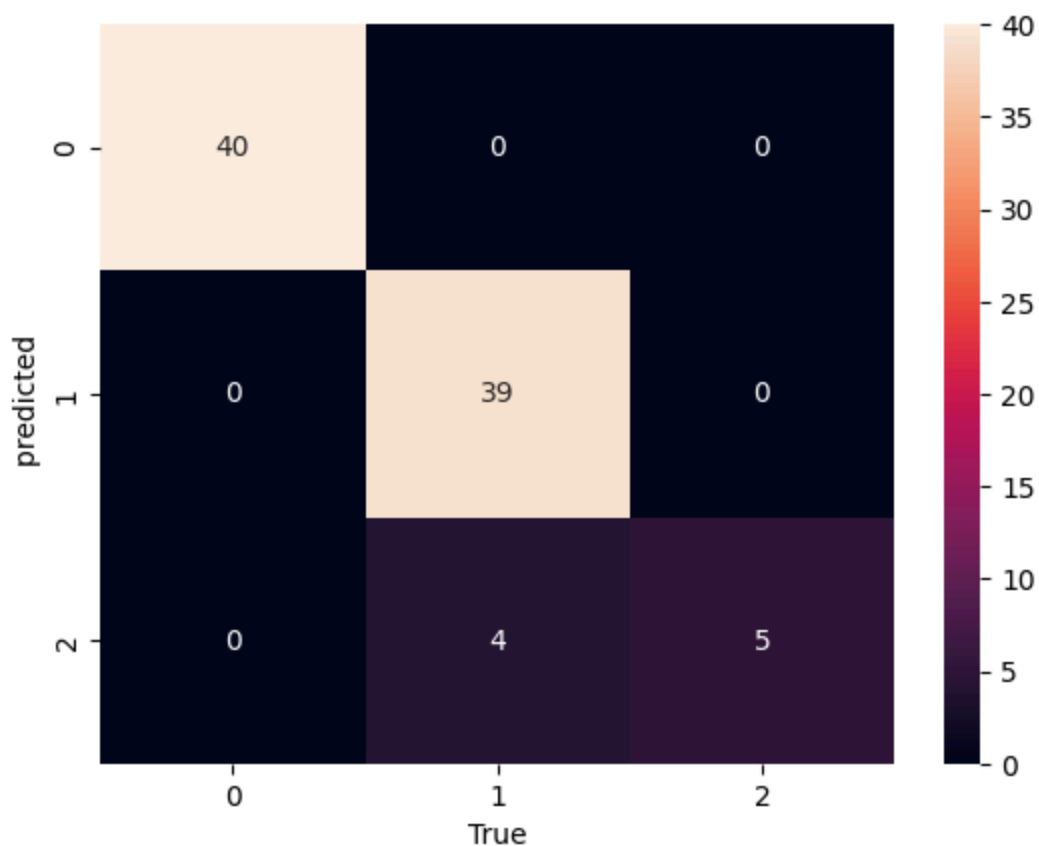
```
In [114... confusion_matrix(y_train, y_pred)
```

```
Out[114... array([[40,  0,  0],  
        [ 0, 39,  0],  
        [ 0,  4,  5]])
```

```
In [115... import seaborn as sns
```

```
In [116... sns.heatmap(confusion_matrix(y_train, y_pred), annot=True)  
plt.xlabel("True")  
plt.ylabel("predicted ")
```

```
Out[116... Text(50.72222222222214, 0.5, 'predicted ')
```



In [117... y

```
Out[117]: array([0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0, 0, 0, 0, 0, 0, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1,
1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1,
1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2])
```

```
In [118... y2= pd.Series(y)
```

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In [119... y2.unique()
```

```
Out[119]: array([0, 1, 2])
```

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In [120... y2[y2==2].shape
```

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Out[120]: (10,)
```

In []: